

Data availability in support of
“The Heterogeneous Surface of Asteroid (16) Psyche”
by Cambioni, S., de Kleer, K., and Shepard, M.
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This folder contains data and results in support of the findings of Cambioni et al. 2022 JGR: planets (in review). Below is a description of the content of the folder.

1. **Readme.pdf:** this readme document
2. **DEMO.ipynb:** Jupiter notebook with examples of how to read and plot the data and results. Versions: python - 3.8.3; matplotlib – 3.2.2.; numpy – 1.19.5
3. **LatitudeArray.npy:** latitude array, dimensions Nx1
4. **LongitudeArray.npy:** longitude array, dimensions Mx1
5. **AverageBrightnessTemperature_Data.npy:** python array of dimension NxM, whose elements are the time-averaged brightness temperature as a function of latitude and longitude corresponding to ALMA data published in de Kleer, Cambioni, Shepard 2021
6. **AverageBrightnessTemperature_GlobalModel.npy:** python array of dimension NxM, whose elements are the time-averaged brightness temperature as a function of latitude and longitude corresponding to the best-fit surface model in de Kleer, Cambioni, Shepard 2021
7. **AverageBrightnessTemperature_GlobalResiduals.npy:** python array of dimension NxM, whose elements are the time-averaged residual brightness temperature (that is, time-average of the model minus the data) as a function of latitude and longitude corresponding to the best-fit model in de Kleer, Cambioni, Shepard 2021
8. **AverageBrightnessTemperature_LocalModel.npy:** python array of dimension NxM, whose elements are the time-averaged brightness temperature as a function of latitude and longitude corresponding to the best-fit properties for each surface area (this study)
9. **AverageBrightnessTemperature_LocalResiduals.npy:** python array of dimension NxM, whose elements are the time-averaged brightness temperature as a function of latitude

and longitude corresponding to the best-fit properties for each surface area (this study)

10. **ThermalInertia_BestFitValue.npy**: python array of dimension NxM, whose elements are the best-fit thermal inertia as a function of latitude and longitude (this study)
11. **ThermalInertia_Uncertainty.npy**: python array of dimension NxM, whose elements are the uncertainty associated with the best-fit thermal inertia as a function of latitude and longitude (this study)
12. **DielectricConstant_BestFitValue.npy**: python array of dimension NxM, whose elements are the best-fit dielectric constant as a function of latitude and longitude (this study)
13. **DielectricConstant_Uncertainty.npy**: python array of dimension NxM, whose elements are the uncertainty associated with the best-fit dielectric constant as a function of latitude and longitude (this study)
14. **GoodnessOfFit.npy**: python array of dimension NxM, whose elements are the goodness of fit of the local model as function of latitude and longitude (this study)
15. **SurfaceCoverage.npy**: python array of dimension NxM, whose elements are the surface coverage (that is, how many times how many times a surface area was observed by ALMA) as function of longitude and latitude (this study)
16. **AltitudeMap.npy**: python array of dimension NxM, whose elements are altitude of Psyche' surface computed by subtracting Psyche's best-fit ellipsoid to its shape model by Shepard et al. 2021

References

de Kleer, K., Cambioni, S., & Shepard, M. (2021, August). The surface of (16) Psyche from Thermal Emission and Polarization Mapping. *The Planetary Science Journal*, 2 (4), 149. doi: 10.3847/PSJ/ac01ec

Shepard, M. K., de Kleer, K., Cambioni, S., Taylor, P. A., Virkki, A. K., Rivera-Valentin, E. G., . . . others (2021). Asteroid 16 Psyche: Shape, Features, and Global Map. *The Planetary Science Journal*, 2 (4), 125. doi:921 10.3847/PSJ/abfdb